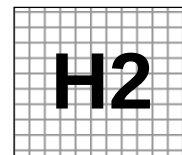


|              |              |                          |
|--------------|--------------|--------------------------|
| Civics Group | Index Number | Name (use BLOCK LETTERS) |
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**ST ANDREW'S JUNIOR COLLEGE**  
**2014 JC2 Preliminary Examinations**

**H2 BIOLOGY**

**9648/3**

**Paper 3: Applications and Planning Question**

Tuesday

16 September 2014

2 hours

Additional Materials: Answer Paper  
Cover Sheet for Section B

**READ THESE INSTRUCTIONS FIRST**

Write your civics group, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

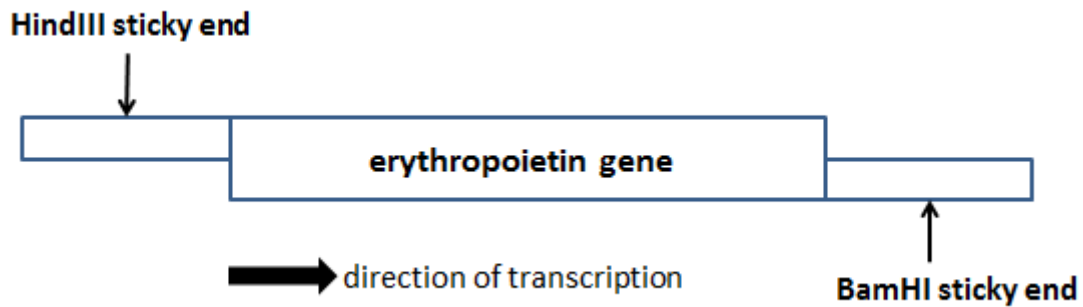
| For Examiner's Use |     |
|--------------------|-----|
| Paper 3            |     |
| 1                  | /14 |
| 2                  | /14 |
| 3                  | /12 |
| 5                  | /20 |
| Total              | /60 |
|                    |     |
| 4<br>(Planning)    | /12 |

This document consists of **18** printed pages.

**[Turn over**

Answer **all** questions.

**1.** The human erythropoietin gene has been isolated from a DNA library. The gene is contained in a HindIII-BamHI fragment (see **Fig. 1.1**). The gene is 2144 bp long and it encodes a 27-amino acid signal peptide and a 166-amino acid mature protein. Erythropoietin is a hormone (glycoprotein) which plays an important role in the production of circulating red blood cells.

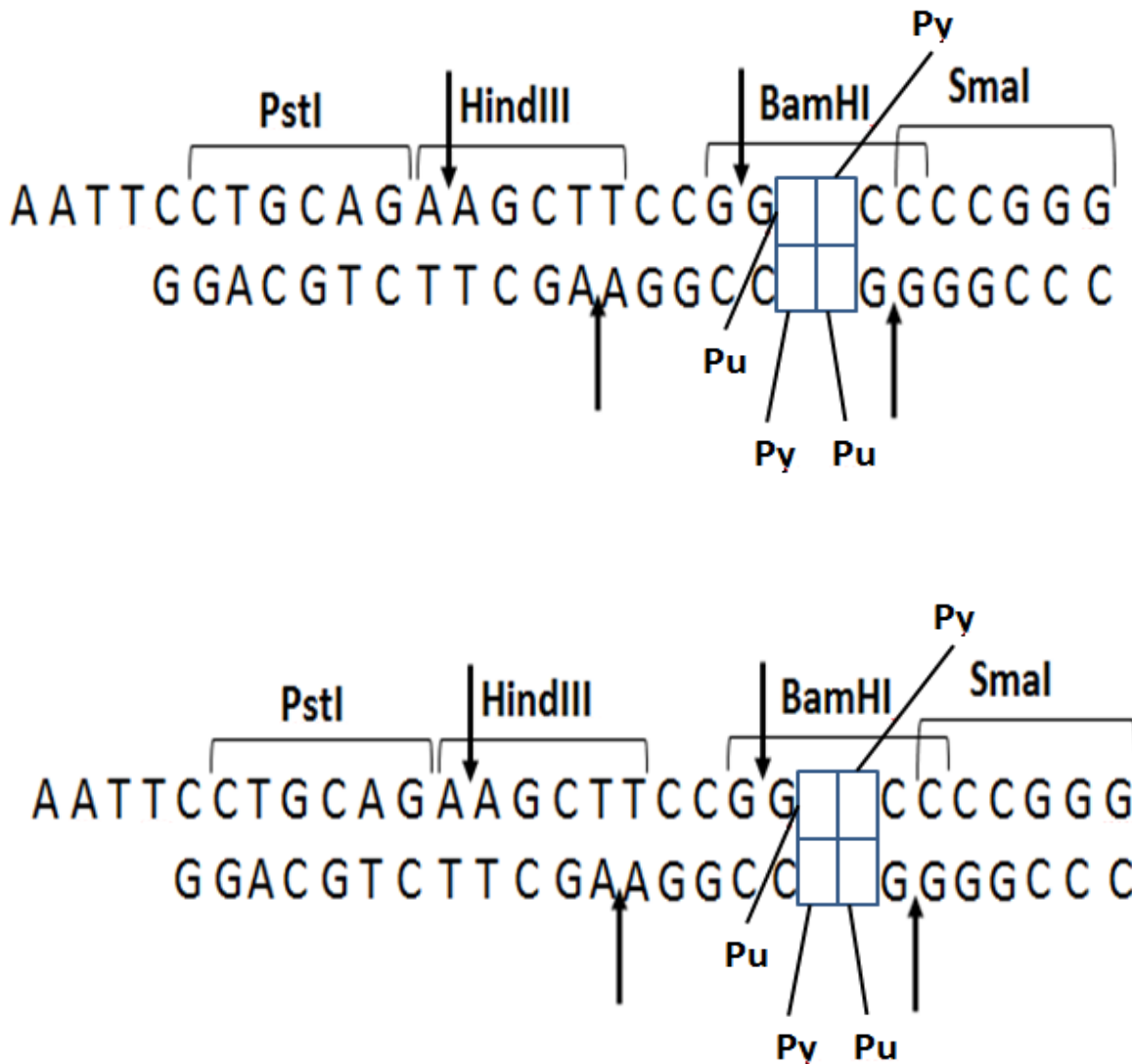


**Fig. 1.1**

**(a)** State the type of DNA library where the erythropoietin gene was obtained from.

.....<sup>[1/2]</sup>

A bacterial plasmid (obtained from *Escherichia coli*), pUC19, with multiple cloning site containing PstI, HindIII, BamHI and SmaI is used as a cloning vector. **Py** may be one of two pyrimidine bases and **Pu** may be one of two purine bases. The DNA bases of the enzyme's target site are shown in **Fig. 1.2**. The cleavage sites are shown by means of arrows.



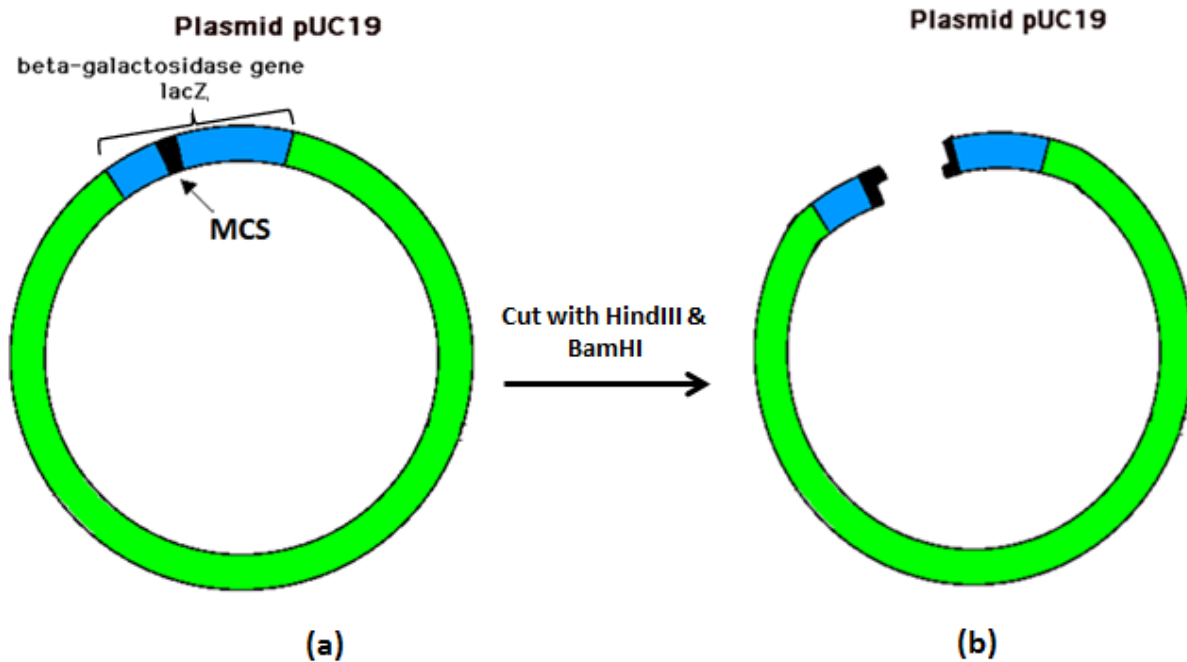
**Fig. 1.2**

**(b)** Complete **Fig. 1.2** to show two possible arrangements of bases of the target site of *BamHI*. [1]

**(c)** Describe the natural function of a restriction enzyme such as *BamHI*.

.....  
 .....[1]

To form recombinant plasmid (pUC 19 with the inserted erythropoietin gene shown in **Fig 1.1**), pUC 19 is cleaved with *Hind*III and *Bam*HI. **Fig. 1.3(b)** shows bacterial plasmid pUC 19 after enzymes cleavage.



**Fig. 1.3**

- (d)** Indicate on the **Fig. 1.3 (b)** the positions of
- lac promoter,
  - the inserted erythropoietin gene showing clearly the position of HindIII and BamHI sticky ends.

[1]

- (e)** Suggest one reason why different restriction enzymes were used to create erythropoietin gene with different sticky ends on each end of this gene.

.....

.....[1]

- (f)** Describe the subsequent steps in the formation of recombinant plasmid.

.....

.....

.....

.....[2]

**(g)** Outline the principle of insertional inactivation of the *lacZ* gene.

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.....

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.....

.....[3]

**(h)** The erythropoietin gene, when introduced in the host cells, *Escherichia coli*, produces erythropoietin that is biologically inactive. Explain why.

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.....[2]

**(i)** It is possible to create an expression vector to carry a eukaryotic gene that can be used within bacteria and yeast cells. State and explain what components are necessary to be incorporate into such vector.

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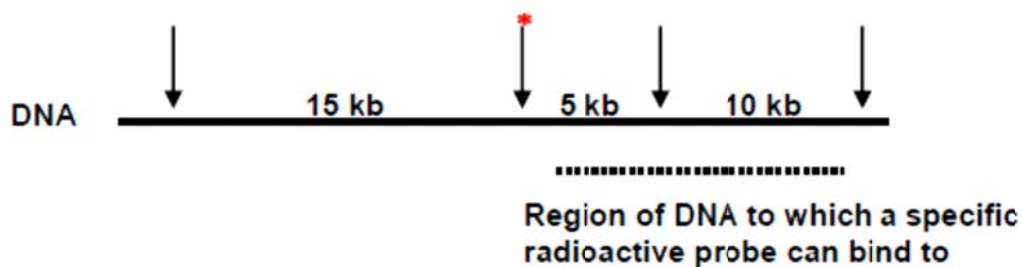
.....[2½]

**[Q1 Total: 14]**

2. People with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because the gene encoding for enzyme Adenosine deaminase (ADA) is mutated.

Genetic screening is used to test for the presence of recessive alleles which cause ADA SCID. Cell samples are obtained from mouth swaps or from the blood.

A RFLP marker linked to the ADA gene from one of a pair of homologous chromosomes is shown in **Fig. 2.1**. The target sites of *EcoRI* are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).



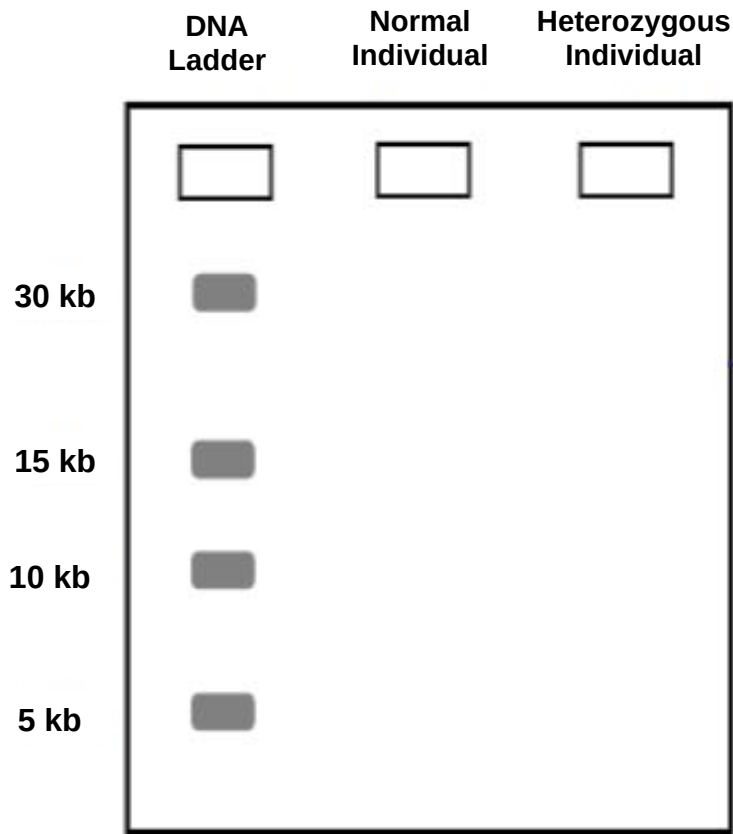
**Fig. 2.1**

A mutation alters one base of the coding sequence at the site marked with an asterisk (\*). DNA from two individuals, one who is normal and the other who is heterozygous for this mutation is cut with *EcoRI* and the DNA fragments separated by gel electrophoresis.

(a) On **Fig. 2.2** below,

(i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individuals tested and label the size of the fragments produced on the gel. [2]

(ii) Circle the fragments which will combine with the radioactive probe and appear as black bands on an autoradiography. [1]



**Figure 2.2**

(iii) Suggest why cells from mouth swabs or blood samples are used rather than gametes.

.....  
 .....[1]

**(b)** Explain how the lack of a functional gene coding for ADA leads to SCID.

.....

.....

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.....

.....[3]

**(c)** Explain why a person without SCID can have the diseased RFLP marker.

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.....[1]

Some patients with SCID have been treated with blood stem cells.

**(d)(i)** Explain how the properties of blood stem cells allow them to be effectively used to fight SCID.

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.....[3]

**(ii)** Suggest why the *ex vivo* method was used in this case.

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.....[1]



**(e)** Gene therapy using the same blood stem cells and retrovirus vectors has succeeded for 15 French baby boys suffering from ADA SCID. However 3 of the little boys later developed cancer and died. Explain how this is possible.

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.....[2]

**[Q2 Total: 14]**

**3.** Environmental stress severely restricts the distribution and productivity of crop plants. Some species of crop plant produce a substance called glycinebetaine (GB) which is a molecule known to protect plants against stress.

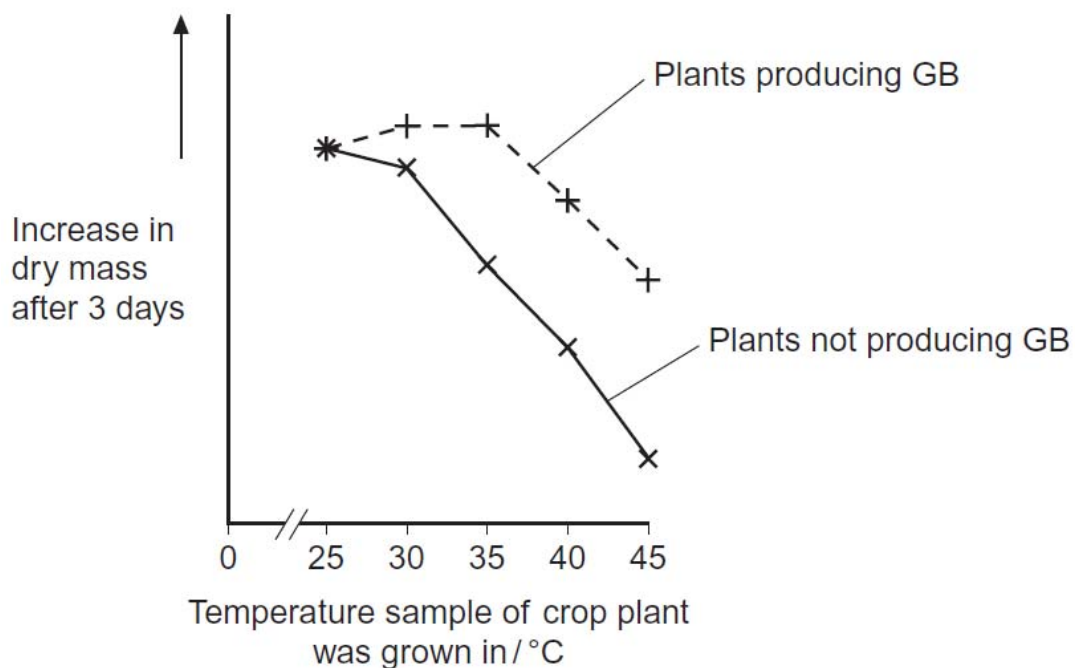
Scientists transferred the gene for GB into a species of crop plant that does not normally produce GB. The plants were then cloned by tissue culture to produce genetically modified plants.

**(a)** Suggest one advantage of cloning the plants by tissue culture.

.....  
 .....[1]

These genetically modified plants then produced GB. The scientists grew large numbers of the same crop plant with and without the gene at different temperatures. After 3 days, they found the increase in dry mass of the plants.

**Figure 3.1** shows their results.



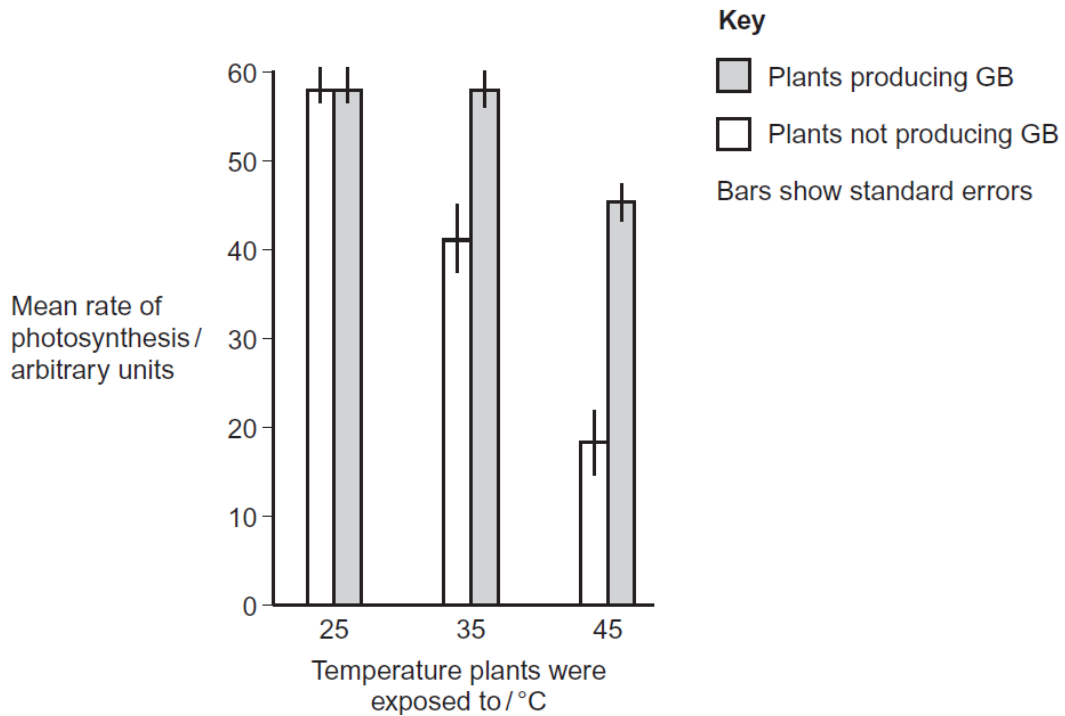
**Fig.3.1**

**(b)** Describe the effect on growth of transferring the gene for GB into this plant.

.....  
 .....  
 .....  
 .....[2]

The scientists measured the rate of photosynthesis in plants that produce GB and plants that do not produce GB at 25 °C, 35 °C and 45 °C.

**Fig.3.2** shows their results.



**Fig.3.2**

**(c)** The scientists concluded that the production of GB protects photosynthesis from damage by high temperatures. Use these data to support this conclusion.

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.....[2½]

**(d)** Use the data from **Fig.3.2** for plants that do not produce GB to explain the effect of temperature on changes in dry mass of the plants shown in **Fig.3.1**.

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.....[3]

A transgenic animal is one that carries a foreign gene that has been deliberately inserted into its genome. The foreign gene is constructed using recombinant DNA methodology. In addition to the gene itself, the DNA usually includes other sequences to enable it to be expressed correctly by the cells of the host.

Atlantic salmon (foreground) which normally grows in Spring and Summer was genetically modified to produce the Aquadvantage<sup>®</sup> salmon (background).



**Fig.3.1**

Credits : <http://foreverconscious.com/wp-content/uploads/2014/04/gmo-salmon-compare.png>

**(e)** Explain why the genetically engineered Aquadvantage<sup>®</sup> salmon (GM salmon) is considered a transgenic animal.

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.....[1½]

**(f)** Describe the effect of the genetic modifications carried out on the GM salmon.

.....

.....[1]

**(g)** Explain the significance of the transgenic GM salmon in solving the demand for food in the world.

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.....[1]

**[Q3 Total: 12]**

#### 4. Planning question

Germination in seeds is a complex physiological process triggered by imbibition of water. As water enters the seeds, their enzymes become activated. Activated enzymes start respiration and use organic matter stored in the seed during respiration to make ATP molecules needed for growth.

You are required to plan an investigation into the effect of temperature on the rate of respiration of germinating seeds.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- Seeds that have been soaked for 24 hours
- Glass beads
- Soda lime
- Syringes
- Glass capillary tubes with bore diameter of 0.4 mm
- Rubber connecting tubings
- Beaker of coloured liquid
- Ruler marked in mm
- Paper towels
- Stop watch
- Electronic weighing balance accurate to 0.1 g
- Incubator

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- describe the method with scientific reasoning so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results table and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

**[Q4 Total: 12]**



[illegible]





**Free-response question**

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5    **(a)**    Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism. [6]
- (b)**    Discuss the goals and benefits of the human genome project. [7]
- (c)**    Discuss the ethical concerns that have arisen from the human genome project. [7]

**[Total: 20]**